



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Department of Computer Science and Engineering DevOps Lab - L3

Lab Assignment No: 5

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Title of Assignment	Jenkins Setup & Freestyle Project

Implementation & Theory

The screenshot shows the Docker Desktop interface. At the top, system resources are displayed: 0.18% / 1600% (16 CPUs available) and 1.06GB / 7.21GB. Below this is a search bar and a toggle for 'Only show running containers'. A table lists running containers:

Name	Container ID	Image	Port(s)	CPU (%)	Memor	Actions
qdrant	9f64d7378874	qdrant/qdr	6333:6333	0%	0B / 0B	[Play] [More] [Delete]
neo4j	d88b7ca7114e	neo4j	7474:7474	0%	0B / 0B	[Play] [More] [Delete]
myjenkins	cc111857475a	jenkins/jen	50000:50000	0.14%	1.06GB	[Play] [More] [Delete]

Below the table is a terminal window showing the following output:

```
6da7907f70f4: Pull complete
Digest: sha256:cce1eeb79902722f3cf7a8831c41e95c39f4a42b9a2981959630cc66539b3c2d
Status: Downloaded newer image for jenkins/jenkins:lts-jdk17
cc111857475a2aa3bb5f0f314f2b7f48614a96f944e0fd76fc2ae87c6cbe2059
PS C:\Users\ASUS> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED          STATUS          NAMES
cc111857475a   jenkins/jenkins:lts-jdk17          "/usr/bin/tini -- /u..." About a minute ago Up About a minut
e               0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp myjen
kins
PS C:\Users\ASUS>
```

At the bottom, system resources are shown: RAM 2.06 GB, CPU 0.12%, Disk 5.09 GB used (limit 1006.85 GB). There is an 'Update' button in the bottom right corner.

docker ps (fresh run) — Started a fresh Jenkins LTS container using Docker with persistent Jenkins storage. Verified that the container was running with ports 8080 (web UI) and 50000 (agent communication) mapped.

```
PS C:\Users\ASUS> docker exec myjenkins cat /var/jenkins_home/secrets/initialAdminPassword
5fd4e68ff0ad4ecb9243c36e352a79a7
```

Initial admin password — Retrieved the auto-generated Jenkins unlock password from inside the running container using docker exec.

Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.

Jenkins 2.541.3

Unlock Jenkins — Entered the auto-generated administrator password on the Jenkins unlock page at localhost:8080 to proceed with the initial Jenkins configuration.

Getting Started

☒ Folders	☒ OWASP Markup Formatter	☐ Build Timeout	☐ Credentials Binding	** commons-lang3 v3.x Jenkins API ** Ionicons API Folders OWASP Markup Formatter ** ASN API ** JSON Path API ** Struts ** Pipeline: Step API
☐ Timestampers	☐ Workspace Cleanup	☐ Ant	☐ Gradle	
☐ Pipeline	☐ GitHub Branch Source	☐ Pipeline: GitHub Groovy Libraries	☐ Pipeline Graph View	
☐ Git	☐ SSH Build Agents	☐ Matrix Authorization Strategy	☐ LDAP	
☐ Email Extension	☐ Mailer	☐ Dark Theme		

Install suggested plugins — Selected the recommended Jenkins plugin installation to install the standard plugins required for Jenkins functionality, including Git integration.

The screenshot shows the Jenkins 'Plugins' page. On the left is a sidebar with links: Updates, Available plugins, Installed plugins (selected), Advanced settings, and Download progress. The main area has a search bar with 'Git' entered. Below the search bar is a table of installed plugins. The table has columns for Name, Health, and Enable. The plugins listed are: Git client plugin (6.6.1), Utility plugin for Git support in Jenkins, Git plugin (5.10.1), GitHub API Plugin (1.330-492-v3941a_032db_2a), GitHub Branch Source Plugin (1983.vfa_27ed961853), and GitHub plugin (1.47.0). Each plugin entry includes a brief description and a link to report an issue. The health status is shown as a green circle with a number inside, and the enable status is shown as a toggle switch.

Name	Health	Enable
Git client plugin 6.6.1	100	On
Utility plugin for Git support in Jenkins	100	On
Git plugin 5.10.1	100	On
GitHub API Plugin 1.330-492-v3941a_032db_2a	97	On
GitHub Branch Source Plugin 1983.vfa_27ed961853	97	On
GitHub plugin 1.47.0	97	On

Verify Git plugin — Checked the installed Jenkins plugins and confirmed that the Git plugin is available and enabled. This provides Git integration required to connect Jenkins with the GitHub repository.

The screenshot shows the Jenkins main dashboard. At the top is the Jenkins logo and a '+ New Item' button. Below that is a 'Build History' section with a table of builds. The table has columns for S, W, Name, Last Success, Last Failure, and Last Duration. The first build is 'Git-Practice-Build' with a status of 'N/A' for both Last Success and Last Failure. Below the table is a 'Build Queue' section with a dropdown menu showing 'No builds in the queue.' and a 'Build Executor Status' section showing '0/2'.

S	W	Name	Last Success	Last Failure	Last Duration
		Git-Practice-Build	N/A	N/A	N/A

Create Freestyle project — Created a new Jenkins project named Git-Practice-Build and selected the Freestyle project type. This project will be used to retrieve the GitHub repository and execute the required build commands.

The screenshot shows the 'Build Steps' configuration for a Jenkins project. The title is 'Build Steps' and the subtitle is 'Automate your build process with ordered tasks like code compilation, testing, and deployment.' Below this is a 'Execute shell' step configuration. The 'Command' field contains the following text: `echo " "`, `echo "Listing files checked out from Git:"`, `ls -la`, `echo " "`, and `echo "Build completed successfully!"`. Below the command field is an 'Advanced' dropdown menu and a '+ Add build step' button.

Git-Practice-Build ▾ / #1 ▾

```
Checking out Revision d8e266951795024ee8c9ba1b63b9cc51758b84b0 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f d8e266951795024ee8c9ba1b63b9cc51758b84b0 # timeout=10
Commit message: "Remove unnecessary source files from assignment folders"
First time build. Skipping changelog.
[Git-Practice-Build] $ /bin/sh -xe /tmp/jenkins3605784137791675920.sh
+ echo

+ echo BUILDING: Git-Practice-Build, BUILD #1
BUILDING: Git-Practice-Build, BUILD #1
+ echo Workspace: /var/jenkins_home/workspace/Git-Practice-Build
Workspace: /var/jenkins_home/workspace/Git-Practice-Build
+ echo

+ echo Listing files checked out from Git:
Listing files checked out from Git:
+ ls -la
total 32
drwxr-xr-x 7 jenkins jenkins 4096 Aug 10 08:04 .
drwxr-xr-x 3 jenkins jenkins 4096 Aug 10 08:04 ..
drwxr-xr-x 8 jenkins jenkins 4096 Aug 10 08:04 .git
drwxr-xr-x 2 jenkins jenkins 4096 Aug 10 08:04 Assignment - 1
drwxr-xr-x 2 jenkins jenkins 4096 Aug 10 08:04 Assignment - 2
drwxr-xr-x 2 jenkins jenkins 4096 Aug 10 08:04 Assignment - 3
drwxr-xr-x 2 jenkins jenkins 4096 Aug 10 08:04 Assignment - 4
-rw-r--r-- 1 jenkins jenkins 1414 Aug 10 08:04 README.md
+ echo

+ echo Build completed successfully!
Build completed successfully!
Finished: SUCCESS
```

Run and verify build — Triggered Build Now and opened the Console Output to verify that Jenkins successfully checked out the GitHub repository and listed the workspace contents. The build completed successfully with Finished: SUCCESS.